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# *The Plug-and-Play Electricity Era*

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**Interoperability to  
Integrate Anything,  
Anywhere, Anytime**

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THE INFORMATION AGE CONTINUES TO TRANSFORM the mechanics of integrating electric power devices and systems, from coordinated operations based purely on the physics of electric power engineering to an increasing blend of power with information and communication technology. Integrating electric system components is not just about attaching wires. It requires the connection of computer-based automation systems to associated sensing and communication equipment. The architectural impacts are significant. Well-considered and commonly held concepts, principles, and organizational structures continue

to emerge to address the complexity of the integrated operational challenges that drive our society to expect more flexibility in configuring the electric power system, while simultaneously achieving greater efficiency, reliability, and resilience. Architectural concepts, such as modularity and composability, contribute to the creation of structures that enable the connection of power system equipment characterized by clearly defined interfaces consisting of physical and cyberlinks. The result of successful electric power system component connection is interoperability: the discipline that drives integration to be simple and reliable.

As machines and automated systems are incorporated into our society, interoperability becomes a necessary capability of networks and devices so that they can provide and receive services and information between each other. The connected parts of the system need to use the services and information that they exchange to operate effectively, in predictable ways, without significant user intervention. When people talk about the modern or smart grid, they

are discussing a concept for which the ease of integration is a necessary foundation (see “Interoperability”). When we think of plug-and-play, we imagine a system paradigm where components come with easy-to-use plugs and sockets that connect and begin to coordinate operations. Too often, our experience falls short of a sense of amazement and reveals the complexity of a world that we would prefer remained out of sight.

### The Parable of the Charging Car

Imagine that Ms. Green purchases an electric car and installs a Level 2 electric vehicle (EV) charger in her garage. Her “smart” unit can respond to signals from her electricity service provider, thus giving her the opportunity to manage her charging cost. The charger requires a 30-A, 250-V, NEMA 6-30 R, 2 P, 3 W receptacle for operation.

Ms. Green hires an electrician to install a receptacle that meets those requirements. She attaches her charger to a wall, plugs it in, and powers it on. The charger searches for a wireless network, and the display instructs her to select an access point. She selects her home network and uses a biomarker (for example, a fingerprint or iris scan) to authenticate the security code. The charger application then asks Ms. Green to select her energy service provider from a list of those in her area and to enter her account number. The charger communicates with the service provider and identifies and registers itself. Ms. Green is prompted to view an email message sent to her by her provider to confirm the registration and select her charging options and savings opportunities. In response, she verifies the arrangement and selects an option for maximum savings; there is no financial penalty if she interrupts charging. The provider and charger exchange information enabling variable-rate charging. At that point, Ms. Green can use the charger.

Note that Ms. Green simply had to authenticate the home network connection and the energy service provider account and then select the options she wanted. She did not have to provide details about the manufacturer of the charger and her vehicle. Figure 1 depicts the steps she needed to take to achieve interoperability with her smart charging system.

For this integration to work easily and reliably, many agreements were put in place and supported behind the scenes. For example, safe connectivity was established using a series of networking security standards. Once the connection was set up, the charger and energy provider traded information that amounted to a handshake, through which the charger identified itself as a specific model. The process for Ms. Green to enter her account number employed an access technique supported by reputable cybersecurity practices, which were also based on standards. Once she confirmed the connection, the provider queried the charger through standards-certified information exchanges. They established the way subsequent information exchanges would occur, configured the interface, and determined the charger’s capabilities through an independent registry (for instance, a database

### Interoperability

ISO/IEC/IEEE standard 24765 defines interoperability as “the ability of two or more systems or components to exchange information and to use the information that has been exchanged.” Interoperability is valuable because it

- reduces the cost and effort for system integration
- improves grid performance and efficiency
- facilitates more comprehensive grid security and cybersecurity practices
- increases customer choice and participation
- establishes industry-wide best practices
- functions as a catalyst for innovation.

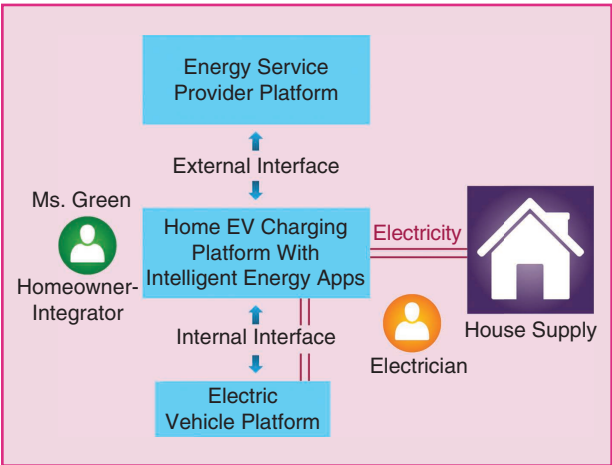


figure 1. Integrating a smart EV charger.

maintained by a consortium of energy service providers and EV charger manufacturers) that described the model's characteristics. With this information, the provider presented Ms. Green with the options for the unit she purchased. She made her choices, and the provider performed any necessary configuration settings. She did not have to engage in a complicated process. Neither did she have to research which functions worked with her charger, call a helpline, or troubleshoot why the different pieces of technology were not working together. She merely had to plug in the charger, establish a network connection, and select her preferences.

### The Lesson: Interoperability Requires Discipline

The desired integration characteristics that result in interoperability are supported through a series of well-documented, tested, and certified agreements, which are mostly embedded in standards that have been developed, adapted, and adopted by a community of stakeholders that reaches well beyond the energy service provider, EV charger supplier, and customer. A term for such a community is an integration ecosystem. The foundational structures that define how the technology components generally fit and interact among each other is the system architecture, and the connection points where the components are integrated are the interfaces (see "Integration Ecosystems").

Methodologies and tools have emerged to help integration ecosystems advance interoperability. The GridWise

Architecture Council's interoperability context-setting framework is shown in Figure 2. It has eight layers grouped into three categories: 1) technical, 2) informational, and 3) organizational. Meshing at an interface requires the integrator to account for alignment in every layer. To illustrate the concepts in this model, consider the way we exchange information through the postal service. For standard mail delivery, the postal service accepts letters that conform to a specified size and weight. We provide the name and address of a recipient along with a ZIP code, return address, and valid postage stamp, all in the right places

### Integration Ecosystems

An integration ecosystem is a community of participating organizations collaborating to address one or more business or social objectives that concern interoperability and ease the deployment of specific technologies. The participants in such communities represent many types of organizations that are needed to support integration objectives. These include managers-owners-users, technology suppliers, service providers, distribution system operators, regulatory and government agencies, consortia/trade associations, and testing-certification bodies. They have an established convening body with champions who drive the group toward alignment in achieving their shared objectives.

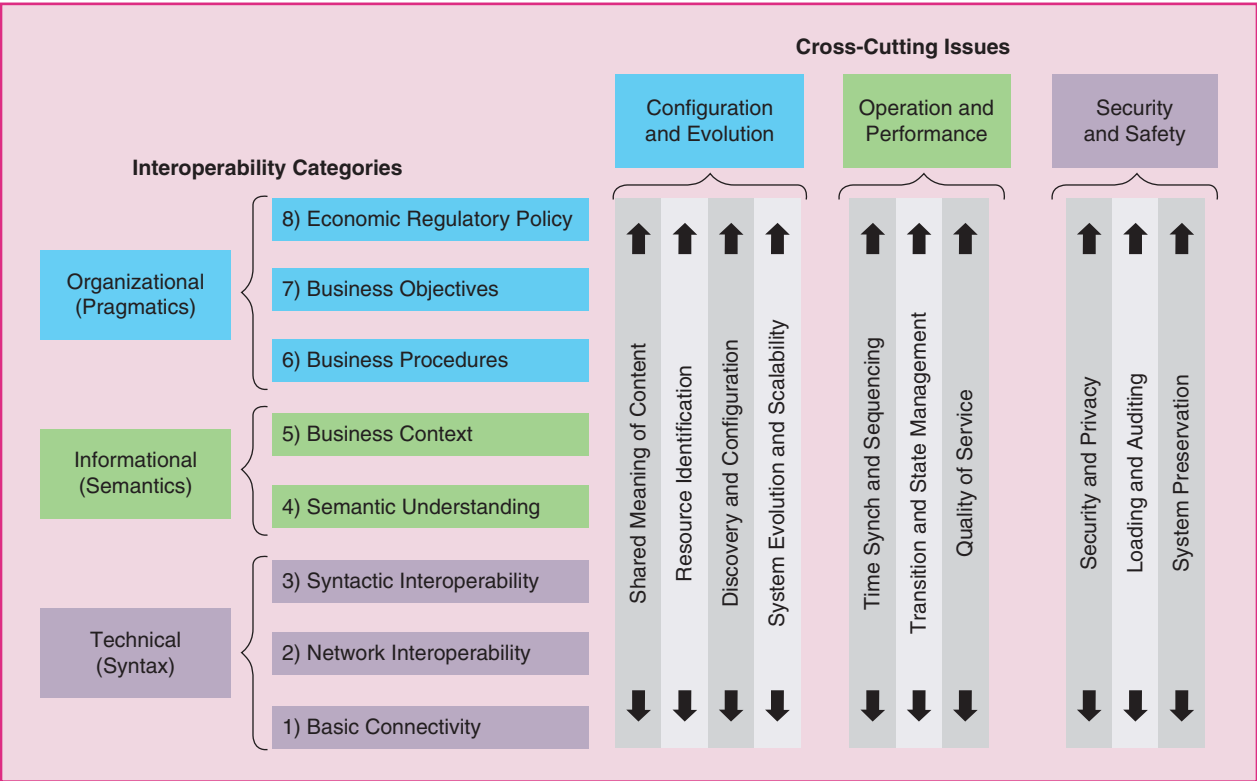


figure 2. The interoperability context-setting framework.

and in a language that can be understood. Once the letter is prepared, it is dropped into a mailbox, where it enters a complex process that involves several hand-offs during its journey. At the recipient's end is another mailbox, where the letter eventually appears and is retrieved. This part of the exchange process represents the technical levels of achieving interoperability. It describes the standards and rules that allow letters to work their way through the postal system. In the information technology world, the communication networks and protocols that allow messages to be routed between application computers provide the analogy. With letters, we rarely give a second thought to this system because the interfaces are simple, and they hide the complex work behind reliable mail delivery.

Yet, agreement on the technical levels only forms a foundation for achieving interoperability. Once the recipient opens the letter, the importance of the information levels increases. Imagine trying to read a letter that is written in a foreign language or uses acronyms with which you are not familiar. Even if the language were familiar, what if jargon associated with an unknown profession was used? Interoperability requires the proper interpretation of a message, and this is represented by the information levels in the context-setting framework. Software engineering uses formal languages to capture the semantics (names of things, meanings, and relationships) in an information model. The business context is also important as information about the same thing may be different depending on its application. Knowledge exchange is not interoperable unless the content of the message can be interpreted. But this still is not enough.

For business information exchange to accomplish an objective (interoperate), the recipient needs to understand what to do with the material. This involves knowing and executing an agreed-upon process. For example, as electricity customers, we agree to the terms and conditions of service. When we receive a monthly bill, we understand that it needs to be paid within a specific period, and we know the acceptable monetary currency and the financial transfer options. We also expect that the utility calculates the bill based on our metered usage. In most situations, the equation for figuring the fee is overseen by a regulator that approves the electric company's tariffs. If we cannot understand the bill, resulting in the wrong amount paid, or if we send our remittance to the wrong place, the process breaks down, and continued service may be compromised. The business processes, alignment of objectives between parties, and regulatory rules are the concerns of the organizational layers in the interoperability context-setting framework. In these layers, the interactions more closely follow human interface agreements, but as our systems become more automated, software is increasingly acting as our agents. As information technology matures, the standards, rules, and codes of operation reflected in the organizational layers are becoming more important (for example, take automated bill paying), and progress in this area is allowing greater opportunities to manage complexity and make integration easier.

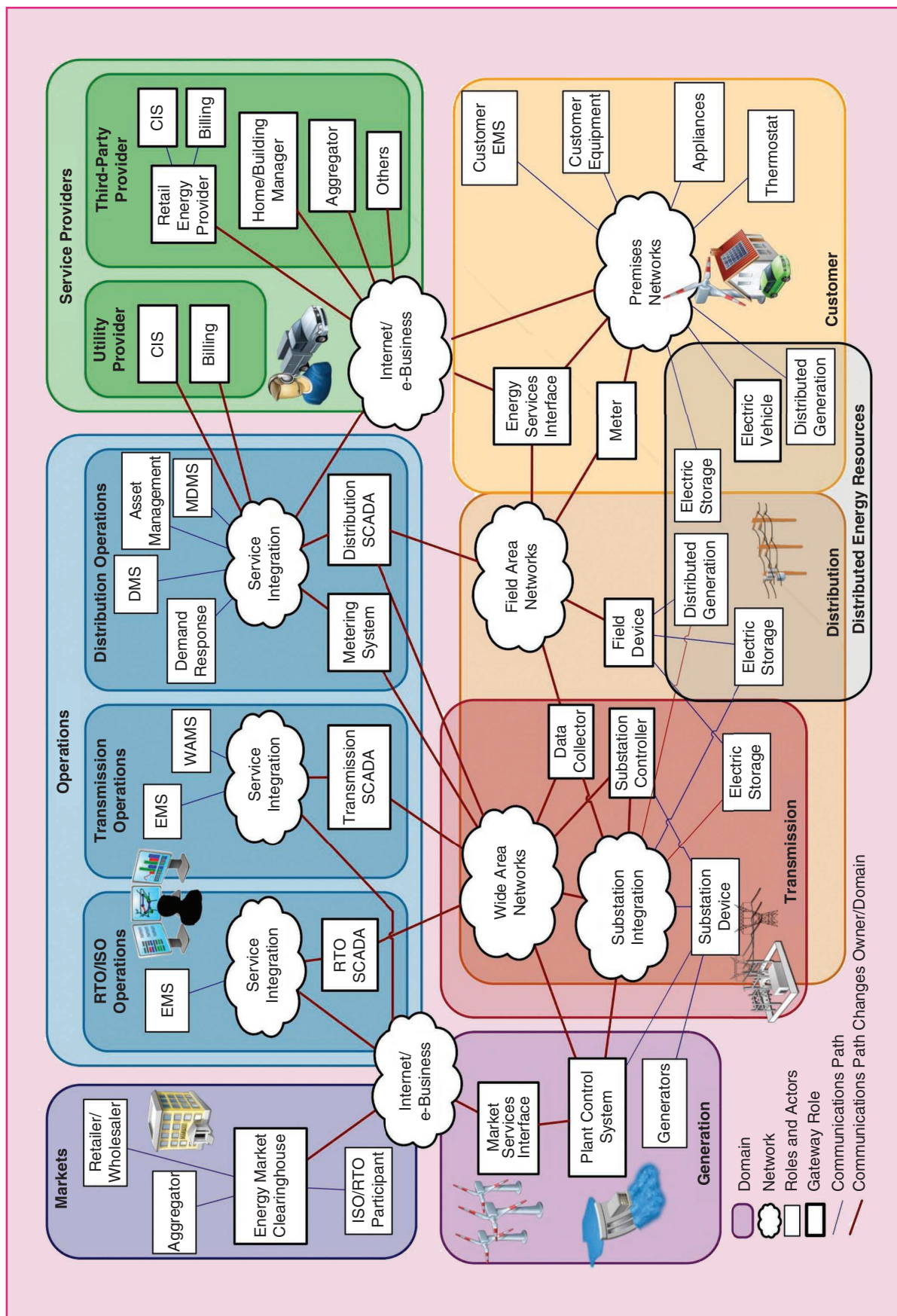
The interoperability context-setting framework helps focus the discussion on categories of interoperability issues, such as the organizational (regulations, policies, contractual agreements, and processes), informational (shared semantics about data, attributes, and relationships between types of information), and technical (communications networks and protocols) as well as cross-cutting issues, such as security, privacy, and safety. All of these concerns must be addressed for interoperability to be achieved effectively.

A core architectural principle for interface specifications is to maintain clear separation and modularity between layers. Cleanly decoupling the capability and functionality associated with the technical (communications) layers and the upper informational and organizational (business) layers' specifications has proven to be particularly important. In many application domains (for example, substation automation, building automation, and EV charging), some standards mix informational layer functionality with specific technical layer protocols. The lack of a layered modularity to distinctly separate agreements about information content from the mechanisms that carry the data inhibits such interface standards from communicating the same material on different protocols. By adhering to the layering principle, the pace of innovation and standards adoption can accelerate by allowing communications protocols, information models, and business processes to coevolve on separate timelines and with different experts involved.

To enable the coordinated operation of a changing mix of transmission and distribution automation technology and the increasing interaction with responsive customer-side systems (flexible loads, generation, and storage), the integration of each component must be simple and reliable. While each function will have specific reasons for grid connectivity and provide or consume different services, general mechanisms that span different technologies are needed to manage integration and operating costs. Because demands are continually changing, the path to advancing interoperability requires continual improvement. Stakeholders in an integration ecosystem must align on a vision for the future, understand the current state of interoperability, identify gaps, prioritize needs, plan steps to meet objectives, and repeat the process periodically to refresh plans.

## **The Reality: A Patchwork of Interoperability**

Interoperability concerns exist in many application domains of electric power system operations. Figure 3 provides a high-level view of the operational spheres, the types of applications within each one, and the interactions (or interfaces) between them (as indicated by the connecting lines). Supporting the ease of integrating these various components is very complex, and it is made even more difficult by areas not covered in Figure 3, such as utility enterprise applications involved with system planning, asset management, and business management (such as billing, personal, and regulatory reporting). The interoperability concepts, challenges,



**figure 3.** The National Institute of Standards and Technology's Smart Grid Framework 3.0, Conceptual Domains for Smart Grid Information Networks. ISO: independent system operator; RTO: regional transmission operator; EMS: energy management system; SCADA: supervisory control and data acquisition; WAMS: wide-area management system; DMS: distribution management system; MDMS: meter-data management system; CIS: customer information system. (Source: U.S. Department of Commerce.)

and tools presented in this article apply generally to information and communication technology integration, although we focus here on operational situations, especially between organizations that must establish agreements.

In the parable of the charging car, the integration scenario involves the energy service provider, the customer's chosen platform, and the EV. This is represented in Figure 3, with interactions among the client, service providers, distributors, and distribution operations. The growing integration and coordination of devices at the customer edge of the system means that millions of interface situations involving EVs, distributed generation, batteries, and flexible loads are at stake. This new world is at our doorstep, and making integration simple and reliable will be a necessary endeavor as our society moves forward.

The degree to which today's distributed energy resources (DERs) are integrated with system operations, such as electric cars, solar panels, and controllable loads, is far from the plug-and-play necessity. Consider the interoperability categories in Figure 2. At the lower technical levels, the aspects of integrating customer-owned devices and systems are addressed by a wide variety of mature standards and support all forms of wired and wireless communication. At the communication protocol level, everything from customers' equipment to their facility management integration uses a wide variety of standards, some open and some proprietary. Each technology ecosystem (such as solar photovoltaics (PVs), EVs, and load management) has its own, often competing, set of protocols for communicating with customer apparatus. Electric utility-oriented projects tend to use technical-level communications standards derived from electric power field equipment automation standards, while customer component suppliers frequently use standards developed in their professional domains (for instance, automotive engineering, industrial systems, and building automation).

Table 1 is based on a U.S. Department of Energy (DOE) grid modernization project that investigated the gaps between DER interoperability standards. The diversity of standards organizations and the types of professional organizations that support them are highlighted across the different types of technologies being integrated. At the interoperability level, the emerging information modeling standards have differences from each DER technology ecosystem (for instance, PV systems, EV charging, battery storage systems, and building-demand response). The format differs across environments, as do information science-based methods and tools, such as the Unified Modeling Language standard, and the knowledge representation for defining models. Some information models are extensions of existing standards that use specialized language formats, making harmonization of specifications across different ecosystems difficult. Existing information models tend to support direct-control types of interactions with DER equipment for reading statuses and setting actions. In this situation, the models must represent all the relevant details of the remotely controlled apparatus so that the algorithms can map the equipment. For example, a system remotely governing a smart inverter on a rooftop solar setup must model all the appropriate characteristics of that equipment,

meaning much information must be exchanged and kept current for the controller to properly do its job.

At the interoperability organizational level, architectural ideas supporting greater separation of decision-making functions that have clear lines of responsibility are emerging in the form of service-oriented requests that describe the desired outcomes independently from how the results are produced. This can promote a business process that significantly simplifies information models based on service definitions and expected performance rather than detailed equipment models that must be exchanged and continually maintained. Standards that uphold service-oriented interactions are starting to be adopted in specific ecosystems.

While there is some progress at the organizational level, there are few grid services available for DER coordination. Most interactions with DER facilities are directly controlled by utilities or third-party aggregators. Service contracts may or may not stipulate the purpose of control, so the grid service or system objective is not always apparent to the DER facility. Some programs are designed to encourage consumer behavior changes in ways that help system operations without using direct-control methods. Examples include tariff schemes, such as demand capacity charges that provide an incentive to stay below a power limit, and time-of-use programs that change the price of energy depending on when it is consumed. Systems in which retail prices are dynamic beyond fixed schedules are rare, but interest in variable rate scheduling is growing.

The following are observations about the interoperability characteristics for aggregators of DERs:

- ✓ Most programs are used primarily for peak capacity management and spinning reserve grid services, but each jurisdiction defines the services differently. The lack of standardization across market systems requires aggregators to customize their technology.
- ✓ Aggregators tend to use proprietary communication and control platforms to directly govern DERs. They work with a customer's facility management system to regulate the local equipment, or they command the DER equipment directly through their own technology. This makes interfaces with DER facilities aggregator dependent and hinders switching between providers.
- ✓ From an architectural framework perspective, there is little coordination between third-party aggregators dealing with wholesale markets, and distribution system operators. Problems in system operation loom as these programs scale to high penetration levels within a distribution feeder.

Cybersecurity issues exist across the organizational, informational, and technical levels as well. Most DER facilities operate without any cybersecurity requirements, although the adoption of standards is improving. Even in California, where the California Independent System Operator (ISO) has set cybersecurity requirements for DER providers, the specifications stop at the aggregator level as DER facilities internally operate without cybersecurity requirements. Existing

cybersecurity-related standards and policies could be useful in addressing secure communications issues. However, the application of these standards remains project specific, since each firm develops its own policies. Much more can be done to develop and circulate best practices and create more commonality that can benefit all organizations.

## A Path Forward

Advancing interoperability is a team sport. Each stakeholder in an integration ecosystem has its own perspective, business drivers, and priorities. The key to reaching agreements involves finding and articulating shared goals that transcend

individual positions. Establishing a vision for interoperability can align stakeholders in a common effort that rises above the struggles of today's integration issues. Regarding the issue of DER integration, the DOE's Grid Modernization Initiative sponsored workshops with diverse stakeholder representation to discuss interoperability concepts and challenges. A strategic vision was molded from those meetings, with the objective of aligning the DER integration community directionally. With a shared vision, a community can assess where it is today, identify gaps between its present situation and desired future, and develop plans to address those shortcomings through deliberate steps.

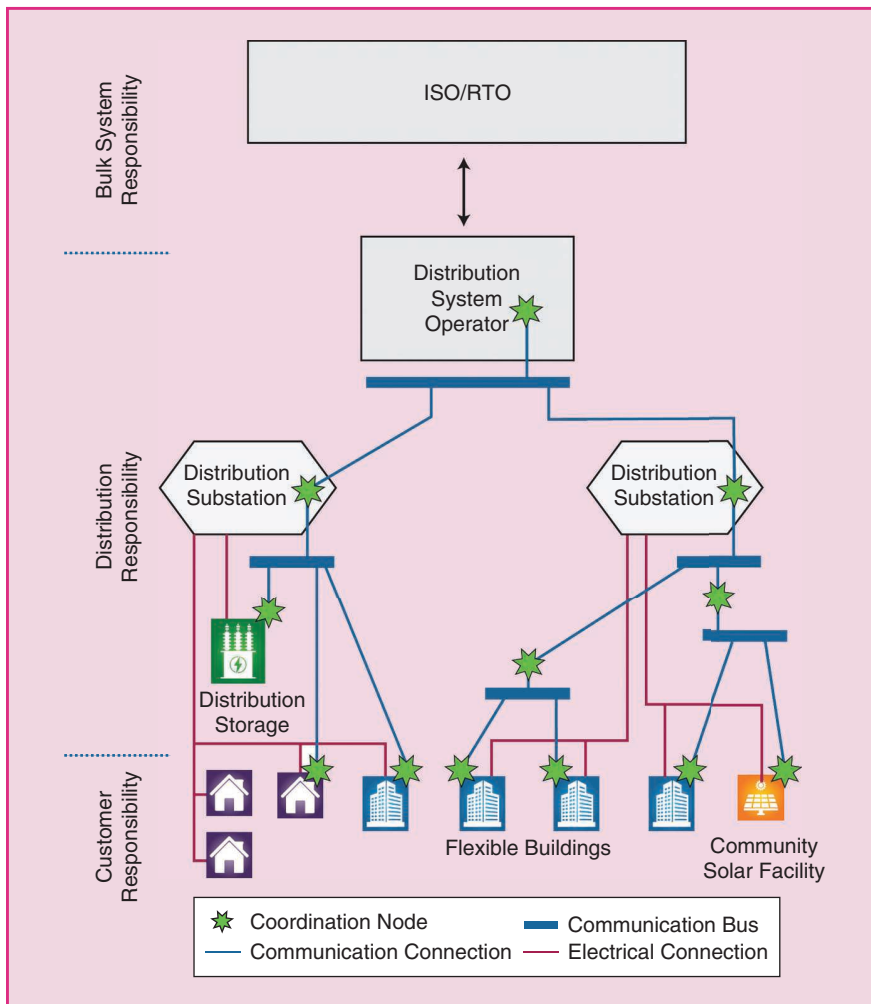
**table 1. An example of interoperability and interconnection mapping across technology domains.**

| Governmentwide Acquisition Contract Stack Layer | Inverter-Based Generation and Storage                                                            | EVs                                                                             | Controllable Loads                                                                                                                | Grid-Connected Microgrids                                         |
|-------------------------------------------------|--------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|
| 8. Economic/regulatory policy                   | Grid service defined by policy or economic market                                                |                                                                                 |                                                                                                                                   |                                                                   |
| 7. Business objectives                          | Grid service enabled as a business objective                                                     |                                                                                 |                                                                                                                                   |                                                                   |
| 6. Business procedures                          | Entity provides signal requesting grid service                                                   |                                                                                 |                                                                                                                                   |                                                                   |
|                                                 | Inverter-based systems/vehicles/controllable loads/microgrid respond to request for grid service |                                                                                 |                                                                                                                                   |                                                                   |
|                                                 | Device controls defined for provision of grid service                                            |                                                                                 |                                                                                                                                   |                                                                   |
| 5. Business context                             | Data modeling relevant for providing a specific grid service                                     |                                                                                 |                                                                                                                                   |                                                                   |
| 4. Semantic understanding                       | SunSpec PV Models<br>SunSpec/MESA Device Models<br>OpenFMB                                       | Vehicle data models                                                             | OpenADR 2.0                                                                                                                       | IEEE P2030.7<br>IEEE P2030.8                                      |
| 3. Syntactic interoperability                   | IEEE 2030.5<br>IEC 61850-7-420<br>IEEE 1815<br>IEEE 2030.2<br>IEEE 1547.3<br>Modbus              | SAE J2847/3 <sup>a</sup><br>SAE J2847/2                                         | OpenADR 2.0<br>ASHRAE 201 (FSGIM)<br>ASHRAE 135 (BACnet-WS)<br>IEC 14908 (LonMark)<br>IEEE 2030.5<br>OBIX<br>OASIS EMIX<br>Modbus | Modbus<br>IEEE 1815                                               |
| 2. Network interoperability                     | TCP/IP<br>ZigBee<br>IEEE 1815<br>CAN bus                                                         | TCP/IP<br>UDP<br>FTP<br>HTTP                                                    | TCP/IP<br>DALI<br>ZigBee                                                                                                          | TCP/IP<br>IEEE 1815<br>CAN bus                                    |
| 1. Basic connectivity                           | Twisted Pair<br>CTA-2045<br>IEEE 802.11 (Wi-Fi)<br>IEEE 802.15.4 (Thread)                        | Twisted Pair<br>CTA-2045<br>IEEE 802.11<br>SAE J1772 (PLC)<br>SAE J2931/4 (PLC) | IEEE 802.11<br>IEEE 802.15.4                                                                                                      | Twisted Pair<br>RJ-45<br>CTA-2045<br>IEEE 802.11<br>IEEE 802.15.4 |
| 0. Interconnection                              | IEEE 1547<br>IEEE 1547.1<br>UL1741<br>IEEE 2030.2                                                | SAE J3072 <sup>b</sup><br>SAE J2894-1 (PQ)                                      | N/A                                                                                                                               | IEEE 1547.1<br>IEEE 2030.7<br>IEEE 2030.8<br>IEEE 1547.4          |

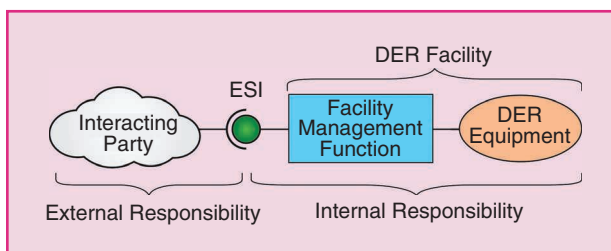
Adapted from Narang in "For Further Reading." An additional level to the interoperability categories was added at the bottom of the table to indicate the electric power interconnection agreements. MESA: modular energy storage architecture; FMB: field message bus; ADR: automated data response; SAE: Society of Automotive Engineers; PLC: power line communications; PQ: power quality; ASHRAE: American Society of Heating, Refrigeration, and Air-Conditioning Engineers; FSGIM: facility smart grid information model; BACnet: building automation and control network; WS: web service; OBIX: Open Building Information Xchange; OASIS: Organization for the Advancement of Structured Information Standards; EMIX: Energy Market Information Exchange; TCP: Transmission Control Protocol; IP: Internet Protocol; CAN: controller area network; UDP: User Datagram Protocol; CTA: Consumer Technology Association; RJ: registered jack; DALI: Digital Addressable Lighting Interface.

<sup>a</sup>SAE J2847/3 supports IEEE 2030.5.

<sup>b</sup>SAE J3072 enables conformance to IEEE 1547.1.



**figure 4.** The utility architecture view, with layered decomposition coordination (see Taft in “For Further Reading”).



**figure 5.** The components of DER and grid integration.

To provide context for the strategic vision, the architectural principle of layered decomposition was applied as an electric power system framework for coordinated operation of DERs. The basic approach involves breaking down system coordination objectives into a master problem and several subproblems, which are solved iteratively until they converge. Each subproblem may be dismantled recursively into sets of subordinate issues. Each level of nested subproblems can be considered a coordination domain. Figure 4 depicts a layered decomposition view of power system coordination. At the

top is the regional transmission coordination domain, which interacts with the distribution system operation coordination domain. In turn, the distribution system operation coordination domain interacts with the primary distribution substation domain, which can interact with further decomposed domains until, eventually, the DER coordination domain is reached.

The principle of layered decomposition informs the identification of the interfaces in the system architecture where problem domain responsibilities can be assigned, and coordination between these areas can be specified. The interfaces become the focus of interoperability agreements. Interactions that skip layers of the coordination framework structure can lead to less optimal or unstable behavior. For example, a market provider that aggregates responsive equipment for interactions with ISO energy markets cannot reliably synchronize with subproblems at the DER facilities level if the coordination frameworks of the affected distribution system operators are not also part of the process. How else would such a solution process know if a

distribution system delivery constraint was violated?

The concept of an energy services interface (ESI) is founded on the principle of layered decomposition. An ESI is a bidirectional, service-oriented, logical interface that supports the secure communication of information between entities inside and outside of a customer boundary to facilitate various interactions between electrical loads, storage, and generation within client facilities and external bodies. Imagine a DER facility as defining the boundary of a customer’s devices and systems. To be grid responsive, it uses automation to manage the equipment and network’s operation. The DER facility, hardware, and system interact by supporting an ESI to external parties. Figure 5 provides a conceptual view of an ESI. The figure introduces the concept of a facility management function to generalize the idea that DER equipment has an intelligent coordination aspect. This allows the decoupling of integration (and interoperability) concerns between stakeholders involved in internal interactions (for example, equipment and facility controls suppliers) and external connections

(for instance, facility controls suppliers, distribution system operators, and market service aggregators).

The facility management function could be as simple as a communicating controller embedded in the smart inverter of a battery or as complex as a management system for a large commercial building, manufacturing facility, or EV-charging parking lot with PV arrays. To interact with DER equipment, the facilities management function coordinates internal interfaces with the hardware. To synchronize with external parties, it supports the ESI.

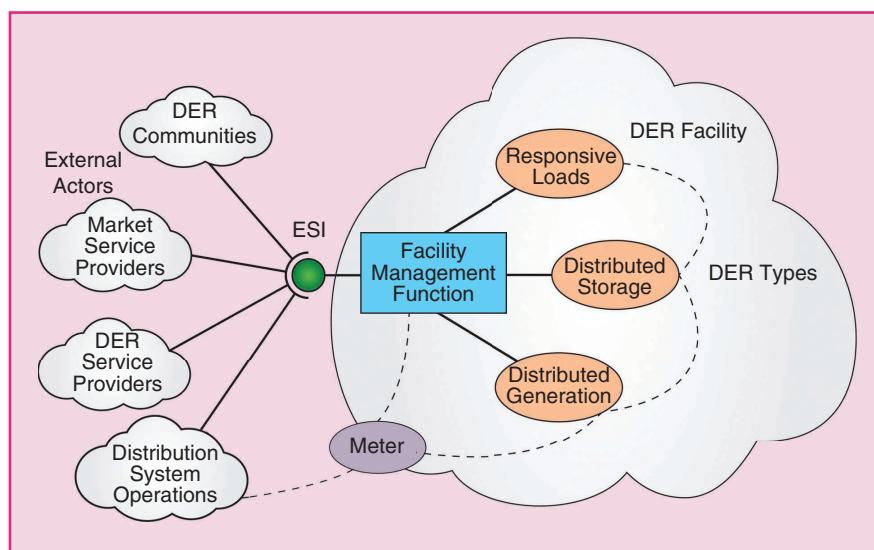
The ESI definition includes the notion of being service oriented. This is a software-systems engineering approach that emphasizes what is expected (the service) and separates or hides how that service is performed. In the case of integration, it means that external parties do not need to know how the DER facility manages its equipment if the service satisfies the terms of the agreement (for instance, performance measures). Much of today's computer software uses the service-oriented design paradigm to emphasize separation of responsibilities between interacting components. By defining and providing access to a service, a program offers a way for its user to accomplish a specific task. A provider stipulates the nature of a service in a functional description, its inputs and outputs, its performance guarantees, and important operational details, such as what happens if it cannot complete the task (for example, an error is encountered). The user, or "caller," provides the required input parameters and receives the outputs, including the anticipated work to be accomplished. Importantly, the user does not know how the service is supported. That is the responsibility of the provider. In this way, the provider may change and improve the way the work is performed without any change to the service interface. That separation of concerns improves interoperability because the user's side of the connection is unaffected.

By expressing grid services using a service-oriented paradigm in an ESI specification, a DER facility can provide the same service for any equipment at its disposal if it satisfies the inputs, outputs, and performance guarantees. This approach provides a focus for conventions, guides, and standards that simplify integration of DER facilities with the grid so that their interactions with external parties are the same regardless of the type of operation. It also provides a buffer to separate integration issues associated with the DER facilities' internal devices, which is important because those situations usually have different requirements and involve a different community of people than interface problems with external parties.

Figure 6 expands the basic concept of a DER facility to encompass one or more types of DERs with some sort of facilities management function. This system can interact with external parties using an ESI. The classes of interacting parties are indicated by the clouds connected by communication paths, on the left, to the ESI. The types of DER equipment are indicated by the ovals connected by different communication paths, on the right. Another important concept in Figure 6 is the meter. To reconcile the resulting actions of a DER facility with the coordinating agreements, sensing and measurement systems need to be in place. The meter monitors the flow of energy between the facility and the electric system. A dotted line is the communication path that connects the meter to distribution system operations, which is the authority responsible for reliable operation. The solid and dotted lines represent integration interfaces for discussing interoperability issues.

### Cultivating a Culture of Interoperability

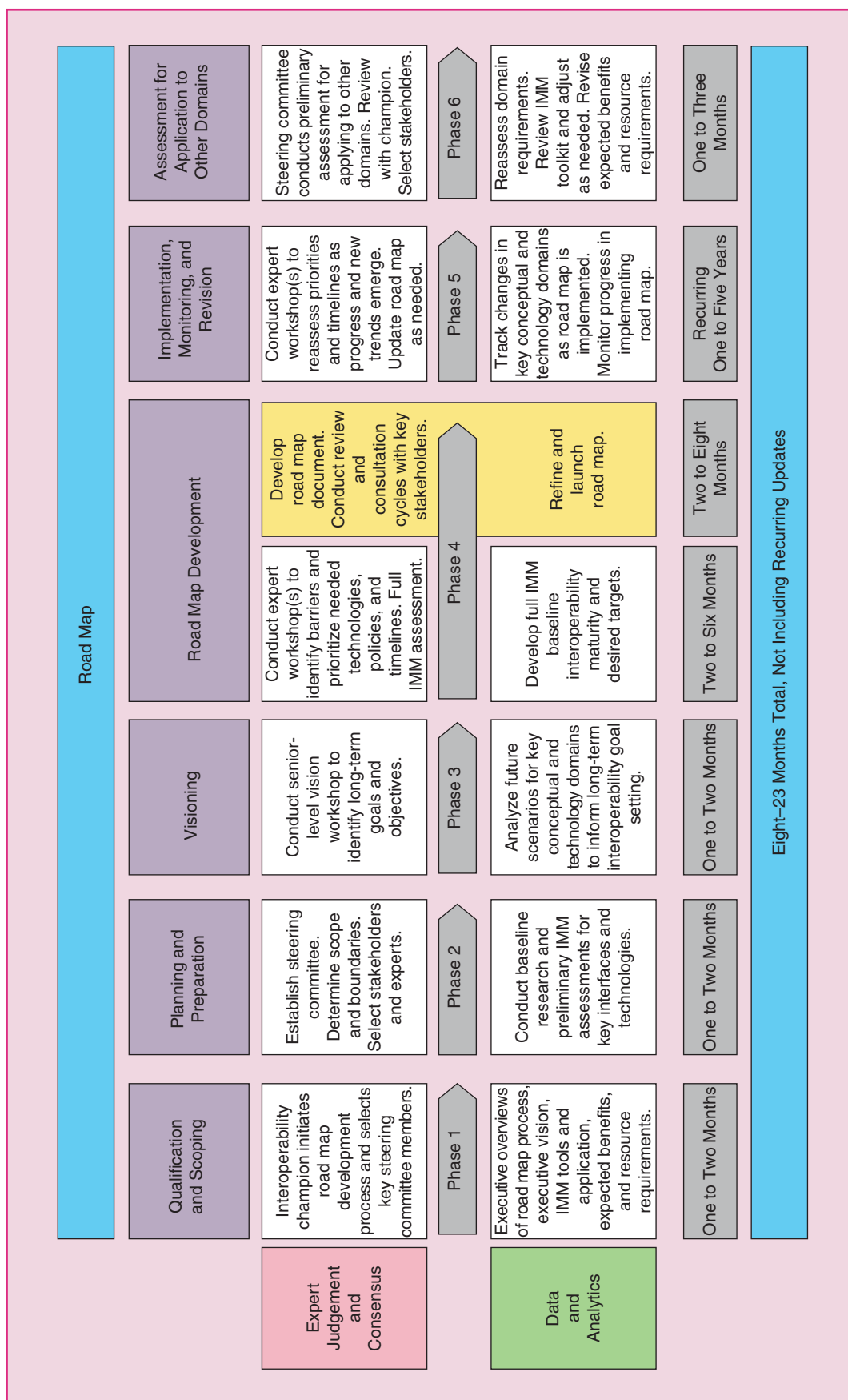
To achieve plug-and-play integration no matter what type of DER system, a cross-technology coordination strategic vision for grid connection can drive the convergence of concepts and approaches. This can lead to the consolidation of interoperability standards and conventions that are adopted and used in deployments by electric system system operators, DER aggregators, and DER facility managers. The strategy must acknowledge that real progress toward adoption requires action in integration ecosystems, where value propositions are tied directly to the process of aligning stakeholders. Table 2 lists strategic activities to advance interoperability for grid modernization and DER integration. A set of general actions for the grid transformation community is included, followed by activities organized using the categories and cross-cutting issues described by the interoperability context-setting framework.



**figure 6.** The visualization of a DER facility conceptual model.

**table 2. Strategic activities.**

| Area                        | Activity                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Community                   | <p>Interoperability vision: Publicize the definitions, concepts, and architectural structures to support the integration vision (for instance, layered decomposition coordination frameworks and ESI concepts for DER facilities' external interface interactions).</p> <p>Communities and ecosystems: Identify communities and ecosystems, and plan to engage them using the road map methodology and tools for defining states, identifying gaps, and prioritizing steps forward.</p> <p>Encourage an interoperability culture: Develop initial pro-forma procurement language to encourage decision makers to fund interoperable investments. Make this readily available to the greater DER community. Sponsor reference implementations with challenges, prizes, and projects that advance the state of the art for interoperability. Recognize champions for interoperability in professional societies and trade groups.</p> <p>Marketing, communications, and education: Develop introductory material for tutorials on interoperability, its benefits, and how to measure its qualities. Plan and support public presentations at conferences and webinars and in publications (for example, technical journals and industry trade magazines).</p> |
| Configuration and evolution | <p>Resource identification: Support unique identification management between all DER facilities and external parties and all equipment within facilities. The support should include a path for legacy devices and systems to be maintained.</p> <p>Registration and discovery: Support registration and discovery mechanisms between all DER facilities and external parties and all equipment within facilities. The support should include a path for legacy devices and systems to be maintained. The mechanisms may initially be ecosystem dependent, with steps to encourage a consolidation of approaches.</p> <p>Scalability: Document requirements and demonstrate scalable approaches for DER integration and coordinated operation.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Security and safety         | <p>Security policies: Work across ecosystems to develop best practices for security policies covering interface specifications for the external and internal interactions associated with DER facilities. This includes methods to quantify the risk associated with the compromise of individual DERs and the aggregate risk arising from the compromise of large numbers of them.</p> <p>Privacy policies: Work across ecosystems to develop best practices for privacy policies covering the interface specifications that are external and internal to DER facilities.</p> <p>Failure mode policies: Support electric system industry initiatives for the high penetration of DERs within grid codes. Support best practices for safe-mode operation during failures in communications and for DER facilities and equipment to honor grid service agreements. Move from DER-type dependencies to generalized DER performance dependencies.</p>                                                                                                                                                                                                                                                                                                          |
| Operation and performance   | <p>Operations and performance characteristics: Document the performance and reliability, error handling, time-ordered dependency, time synchronization, and transaction and state management requirements. These are important for DER facilities to qualify for specific grid services, in defining expectations, and establishing equipment ratings and certifications.</p> <p>Ratings and certification: Establish programs to rate and certify products to common operation and performance specifications for supported grid services requirements across all DER integration ecosystems.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Organizational              | <p>Model interface specification framework: Define a template for interface specifications. The model should cover the specification of how interoperability characteristics are addressed for each DER facility interaction with external parties and internal equipment. This includes the following: the discovery/registration/qualification process, negotiation process for a grid service, operations process, measurement and verification, and settlement and reconciliation.</p> <p>Grid services for interoperability: Establish or adopt a grid services reference model for DER participation with electric system parties.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Informational activities    | <p>DER performance characteristics: Define an information model of the performance characteristics required to qualify for specific grid services.</p> <p>Information modeling: Define the information models employed in the external and internal DER facility interactions using well-recognized modeling language. The information models for different energy technologies should converge to form a representation common to all types of technology to support the ESI concept.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Technical                   | <p>Consolidate protocols: For each ecosystem, develop plans to consolidate (deprecate/reduce) the number of protocols used in new deployments. Ensure that technical communication network layers are defined independently from informational and organizational characteristics.</p> <p>Transition path: Develop transition paths for legacy protocol deployments within ecosystems.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |



**figure 7.** The road map methodology.

## Methods and Tools

Finding ways to measure the complex interoperability dimensions can help integration ecosystems clarify gaps and challenges. Since interoperability requires forming agreements among interacting parties and related stakeholders, the method for addressing concerns follows a continuous process improvement problem in a situation that matures over time. Using a maturity-model approach provides a measurement structure for assessing how a set of characteristics evolves.

An interoperability maturity model (IMM) measures the progress toward integrating the information and communications technology aspects of intelligent devices and systems to coordinate their operation with others in the grid. The structure provided by the interoperability context-setting framework (Figure 2) has been used to organize a set of criteria statements that can be applied to quantify the maturity level of various dimensions of coordinated connection.

Real progress to address interoperability is made by those working to achieve business and policy objectives within ecosystems that generally form to promote technology deployments. The development of standards and related testing material often is performed with a requirements analysis based on markets' needs to deploy technology. The process can be opportunistic, relatively narrow, and somewhat ad hoc. A road map methodology (for example, the process of developing a high-level plan of action) geared to stakeholder engagement can provide perspective and direction to improve interoperability. Figure 7 shows an interoperability road map development methodology. This process has been designed to emphasize stakeholder engagement, with a guiding principle that once the participants reach a consensus on shared goals and results, their relationships can help to support the plan's implementation and increase the likelihood that they will execute it successfully. The IMM tool is used throughout the development process to guide and inform the road map partners through the complex dimensions of interconnectedness as they look to understand the state of interoperability, articulate the gaps and challenges, express desired levels of maturity, and formulate a prioritized set of actions to achieve their goals.

## Summary

Advancing interoperability takes foresight and upfront work on complicated issues, such as clarifying the architectural structure of the system, that are crucial to identifying the interface points and addressing scalability requirements. The supporting standards and guides need to consider approaches that allow the power system to evolve. Integration mechanisms need to be able to accommodate system objectives and technology solutions that will change over time. Underlying every interoperable system is hard work by many people, usually clustered on specific portions of the electric system, as well as business opportunities and tech-

nology expertise (ecosystems). Providing a vision for the future integration of devices and systems can raise awareness across ecosystems and steer parties toward a common direction. With tools to measure the state of interoperability and a methodology for assessing and prioritizing gaps and challenges, stakeholders can create road maps that lead toward generally applied approaches that transcend individual environments and leverage common tactics to support successful deployments.

## For Further Reading

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